

Cover letter — *Nature Communications* submission

Manuscript title: An eight-gene thyroid differentiation–silencing axis stratifies radioiodine response across independent cohorts.

Corresponding author: Seungho Cook · Seoul National University Bundang Hospital · kukshomr@gmail.com · 2026-05-21.

Dear Editor,

We submit for your consideration the manuscript "An eight-gene thyroid differentiation–silencing axis stratifies radioiodine response across independent cohorts" for publication in *Nature Communications*. Radioactive iodine fails in one of every three patients with distant metastatic differentiated thyroid cancer, and the field has lacked a parsimonious molecular readout that anchors clinical decision-making before therapy. We address that gap with a unified, multi-cohort, multi-pillar validation of an eight-gene differentiation–silencing axis that we propose as a risk-stratification readout — not a stand-alone biomarker — for radioiodine response in thyroid cancer.

Three features of the work make it appropriate for *Nature Communications*. First, the manuscript integrates four independent data pillars (bulk RNA, single-cell RNA, proteome and DNA methylation) over more than 800 tumours and 14,000 single cells, providing the strongest single statistic in the field at proteomic level (ATC vs PTC dark-matter Fisher OR = 8.54, $p = 2.7 \times 10^{-15}$). Second, the eight-gene panel is a strict subset of the canonical TDS-16 used by the TCGA-THCA Cell 2014 paper and lies inside the 64-gene eTDS framework of Boucai 2023; we therefore deliver clinical parsimony — 50% of the gene cost at 99.5% of the discriminative power — without sacrificing field-canonical grounding. Third, we explicitly frame radioiodine refractoriness as a non-binary, gray-zone phenotype using Mu et al.'s 4-class uptake patterns ($n = 214$), and we provide three independent refutations of the canonical cherry-pick concern (leave-one-gene-out, matched-variance permutation null, TDS-16 sensitivity) on the label-anchored cohort.

We have made our claim ladder deliberately conservative. We do not claim the panel "predicts radioiodine response" in a regulatory or biomarker sense — that claim awaits a prospective, multi-centre validation we explicitly outline as the natural next step. We claim, with careful tier mapping, that the panel is *associated with* radioiodine avidity, *stratifies* tumours along an interpretable axis, and is *consistent with* radioiodine refractoriness across discovery and label-anchored cohorts. Limitations (small Tier-2 cohorts, no Tier-1 raw data in hand, an honest direction-reversal in GSE299988 that we attribute to a lymph-node-positive selection confound) are disclosed transparently in the main text.

We believe the manuscript will be of broad interest to thyroid oncologists, endocrine surgeons, molecular pathologists, and the wider radiogenomics community. We have ensured complete data and code availability for independent replication, and we share all per-sample panel scores and zone assignments as supplementary tables.

The work has not been published elsewhere and is not under consideration at another journal. All authors have approved the submission and declare no competing interests.

Thank you for your consideration.

Yours sincerely,

Seungho Cook (국승호) Co-Principal Investigator Department of Internal Medicine, Seoul National University Bundang Hospital ORCID 0000-0000-0000-0000 · kukshomr@gmail.com

with

Yu Hyeong-won Co-Principal Investigator Department of Internal Medicine, Seoul National University Bundang Hospital